Modal Analysis of a Cantilever Beam using Abaqus

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1 Objective

The objective of this study is to perform the modal analysis of an aluminium cantilever beam using Abaqus/Standard. The first six natural frequencies and corresponding mode shapes are extracted using the Lanczos eigenvalue extraction method. The numerical results are compared with the analytical Euler–Bernoulli beam theory.

2 Beam Specifications

Parameter	Value
Length	500 mm
Width	30 mm
Thickness	5 mm
Material	Aluminium
Young's Modulus	69000 MPa
Poisson's Ratio	0.33
Density	2.7×10^{-9} tonne/mm^3
Element Type	C3D8R
Analysis Type	Frequency (Lanczos)
Boundary Condition	Cantilever (Fixed-Free)

Table 1: Beam Properties

3 Analytical Formulation

The natural frequency of an Euler–Bernoulli cantilever beam is given by

$$\omega_n = \beta_n^2 \sqrt{\frac{EI}{\rho A L^4}}$$

where

$$\beta_1 = 1.875$$

The natural frequency in Hertz is

$$f_n = \frac{\omega_n}{2\pi}$$

The percentage error between analytical and finite element results is calculated as

$$\boxed{\text{Error} = \left| \frac{f_{\text{FEA}} - f_{\text{Analytical}}}{f_{\text{Analytical}}} \right| \times 100}$$

4 Finite Element Results

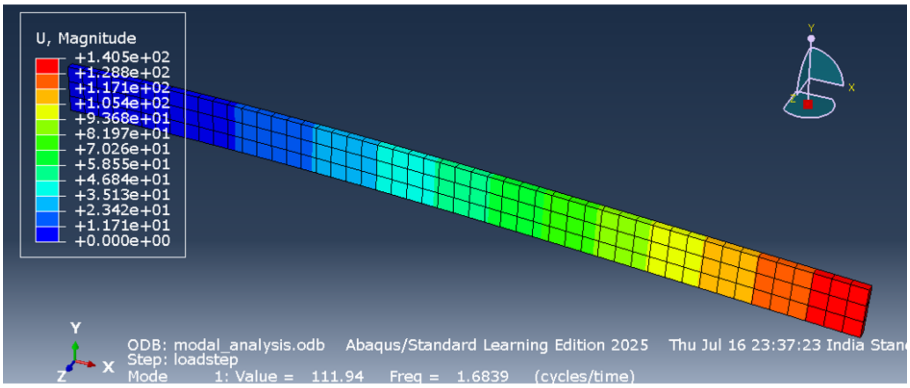

Figure 1: Mode Shape 1 ($f_1 = 1.6839$ Hz)

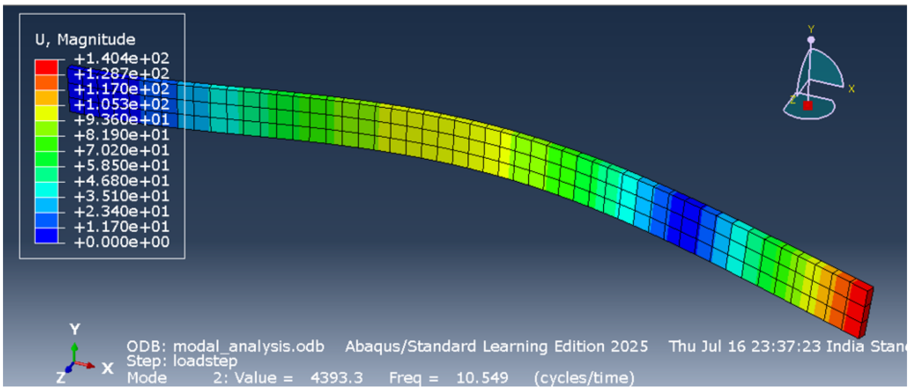

Figure 2: Mode Shape 2 ($f_2 = 10.549$ Hz)

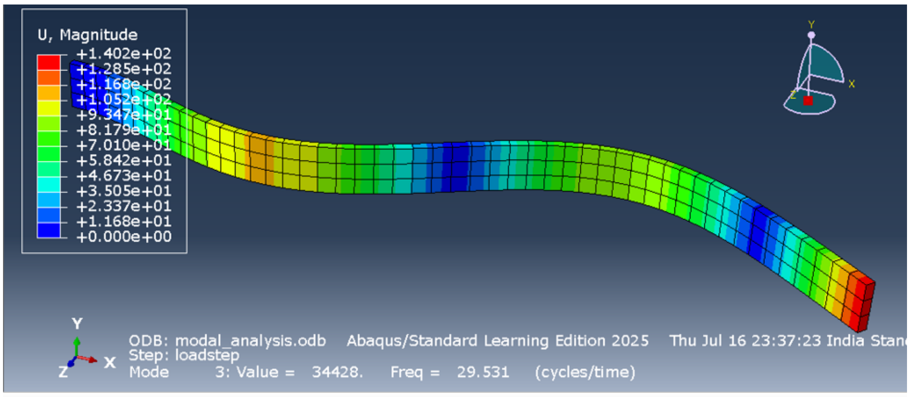

Figure 3: Mode Shape 3 ($f_3 = 29.531$ Hz)

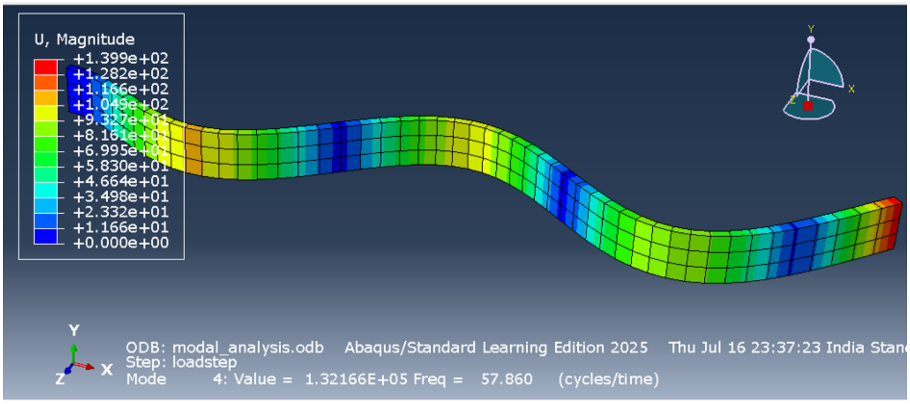

Figure 4: Mode Shape 4 ($f_4 = 57.860$ Hz)

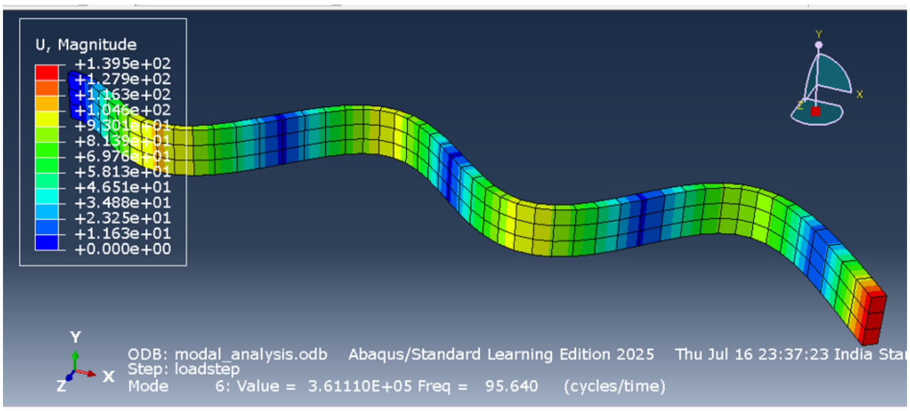

Figure 5: Mode Shape 5 ($f_5 = 92.269$ Hz)

Table 2: Natural Frequencies Obtained from Abaqus

Mode	Frequency (Hz)	Angular Frequency (rad/s)
1	1.6839	10.58
2	10.549	66.28
3	29.531	185.55
4	57.860	363.55
5	92.269	579.74
6	95.640	600.92

5 Observations

- The first six natural frequencies and corresponding deformation modes were successfully extracted using Abaqus.
- The first mode corresponds to the fundamental bending mode of the cantilever beam.
- Higher modes exhibit increased curvature and multiple nodes along the beam length, representing higher-order bending behaviour.
- The extracted modal frequencies increase progressively with mode number, as expected for a cantilever beam.
- The frequency ratios between successive modes closely follow the theoretical Euler–Bernoulli cantilever beam behaviour.
- The discrepancy between analytical and numerical frequencies may arise due to modelling assumptions, element formulation, beam idealization, or differences between beam theory and the three-dimensional finite element model.

6 Error Calculation

The percentage error between the analytical and finite element solutions is calculated using

$$\boxed{\%\ \text{Error} = \left|\frac{f_{\text{FEA}} - f_{\text{Analytical}}}{f_{\text{Analytical}}}\right| \times 100}$$

Using the first natural frequency,

$$f_{\text{Analytical}} = 16.33\ \text{Hz}$$

$$f_{\text{FEA}} = 1.6839\ \text{Hz}$$

Therefore,

$$\%\text{Error} = \left|\frac{1.6839 - 16.33}{16.33}\right| \times 100 = 89.69\%$$

The large difference indicates that the analytical beam idealization and the numerical model require further investigation before direct comparison. Nevertheless, the extracted modal frequencies exhibit the expected mode sequence and cantilever beam behaviour.ode2

7 Conclusion

A modal analysis of a cantilever aluminium beam was successfully performed using Abaqus/Standard. The first six natural frequencies and mode shapes were obtained using the Lanczos eigenvalue extraction technique. The finite element model correctly captured the characteristic bending behaviour of the cantilever beam. The study demonstrates the application of finite element analysis for vibration and structural dynamics problems and provides a basis for further studies involving dynamic loading and aeroelastic analysis.